The Agony and the Ecstasy

Constructing a "Crash-Filtered" Equity Index using Machine Learning

Tristan Leiter

Vienna University of Business and Economics

Supervisor: Prof. Kurt Hornik

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The Agony and Ecstasy

Passive indices are exposed to both extremes

Long-run index returns are driven by a few high-performing stocks, while a considerable share of constituents suffer catastrophic declines.

Ecstasy

- Extreme winners carry a large share of long-run wealth creation.
- Winners can be volatile, expensive, or distressed.
- Thus, even a small amount of disregarded winners can hurt performance.

Agony

- Some constituents experience deep and persistent drawdowns without recovery.
- Passive strategies may hold imploding firms for too long.

Solution

The challenge is not to eliminate highly volatile stocks. It is to distinguish recoverable volatility from persistent implosion. The empirical question is whether catastrophically declining firms are forecastable using ideas from credit-risk modelling, but with market-based distress signals.

Research Question

Main research question

To what extent does a “Crash-Filtered” equity index, constructed via probabilistic implosion modeling, generate superior risk-adjusted returns compared to the market benchmark and traditional minimum-volatility strategies?

The probabilistic implosion model follows the Catastrophic Stock Implosion approach proposed by Tewari et al. (2024). Instead of relying only on formal bankruptcy declarations, CSI uses market-based price-path signals.

Subquestions:

- How does the integration of Autoencoders for feature engineering impact the Average Precision of Ensemble models compared to those trained solely on raw financial data?
- To what extent do Ensemble methods reduce the False Positive Rate (classifying recoverable volatility as CSI) compared to traditional volatility-based exclusion strategies, while maintaining Recall?
- Which features are most important for distinguishing between "Zombie" firms (CSI) and non-zombie firms?

Dataset

Empirical design

The final panel keeps only observable CRSP firm-years: each row has the required monthly return and market-cap scaffold. Temporary and permanent CSI use the same 188,460-row scaffold; unresolved $y = \mathrm{NA}$ rows are retained in the panel but excluded from supervised model metrics.

Data sources

CRSP
monthly returns, market cap, delistings, wealth paths

Compustat
annual accounting fundamentals

FRED
macro variables

Universe

Minimum size
market cap $\geq$ USD 100M

Observable scaffold
188,460 annual firm-year rows in both tracks

Index buckets
Total, Large, Mid, Small Cap

Sample period

Coverage
Jan 1993 – Dec 2024 (32y, monthly frequency)

Supervised labels
train/test labelled; OOS may retain unresolved labels

metrics use labelled rows only

Temporal split

Train/CV	Test	OOS
143,173 rows	18,111 rows	27,176 rows

1993 — 2015 — 2019 — 2024

Methodology I: Response Variable Classification

Following Tewari et al. (2024), a stock is classified as a CSI candidate if it satisfies all three path criteria:

- **Initial Crash (C):** more than a C drawdown from the trailing peak marks the beginning of the "zombie" period. $C \leq -80\%$
- **Non-Recovery (M):** maximum cumulative return remains below the recovery ceiling M during the zombie period. $M \leq -20\%$
- **Zombie Period (T):** duration of the confirmation window during which non-recovery must hold. $T = 18$ months

The prediction task is annual. A trigger in calendar year $t+1$ gives the firm-year observation at t a positive label when the track-specific confirmation rule is met:

$$y_{i,t} = \begin{cases} 1, & \text{if stock } i \text{ triggers a qualifying CSI event within } [t, t+h], \\ 0, & \text{otherwise.} \end{cases}$$

Applying the paper classification

By strictly applying the methodology proposed by **?**, one finds 8,517 response observations for the selected CRSP-dataset within the 1993 to 2024 time period. The prevalence of the response variable is similar across the Train and Test datasets. Overall, the whole dataset has 188,460 observations, with roughly 75% being clustered in the Train set.

Split	Obs. rows	$y = 0$	$y = 1$	Labelled	Prev.
Full	188,460	171,269	**8,517**	179,786	4.74%
Train	143,173	136,630	6,543	143,173	4.57%
Test	18,111	17,307	804	18,111	4.44%
OOS	27,176	17,332	1,170	18,502	6.32%

Motivation for robustness

Even though the paper outlines that a combination of classification grid-values were tested, the selected combination still appears arbitary. For this reason, further robustness checks will be conducted to evaluate the chosen grid-values. It will be asked whether the dataset actually contains the bankrupt firms and how different the prevalence-rate of the response variable is when testing different grid-combinations.

Robustness I: Do CSI-firms go bankrupt?

Validation question: Among firms that eventually delist or bankrupt, how many had already been classified as CSI?

Scenario	CSI rows	CSI firms	CRSP 572-574 firms	Detected	Missed	Detection rate
Before: `confirmed_csi` only	8,369	5,642	629	151	478	24.0%

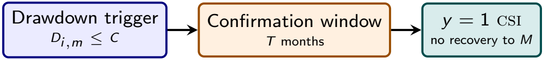
CRSP bankruptcy-related delisting codes 572, 573, and 574 are used as the terminal-distress audit set.

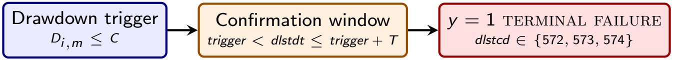
Core issue: The low-detection rate causes methodological issues. If one sticks strictly to the paper, then the CSI methodology should catch firms which do not recover anymore. Yet, the data shows a different result.

Methodology II: “Temporary-CSI”

Revisions to the classification methodology

The original CSI-classification grid remains unchanged. It is proposed to explicitly add all firms, which match the CRSP-codes 572,573 and 574, to the dataset after observing a valid drawdown trigger. This classification methodology will henceforth be referred to as ”Temporary-CSI”.





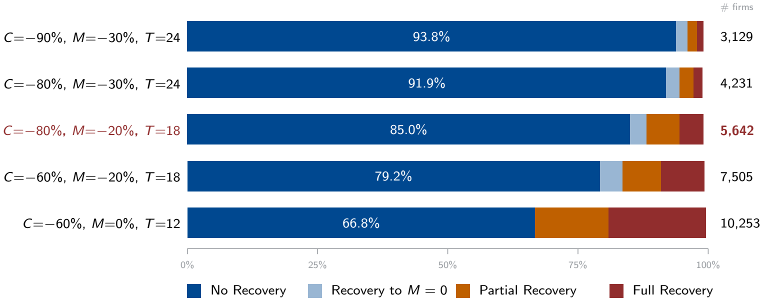
Scenario	Bankrupting	Detected	Missed	**Detection**
Before: `CSI` only	629	151	478	24.0%
After: `CSI` + terminal failure	629	**545**	84	**86.65%**

The 84 remaining misses are treated as methodological exclusions under the retained CSI-trigger rule, not as known pipeline errors.

Robustness II: Do CSI-classified firms recover again?

In addition to bankruptcy checks, robustness checks were also conducted to determine whether CSI-classified firms are in persistent decline.

A grid-search for the three classification paramaters was conducted to track the five-year recovery of firms after classification. Subsequently, each firm is sorted into one of five buckets depending on its five-year performance after classification.

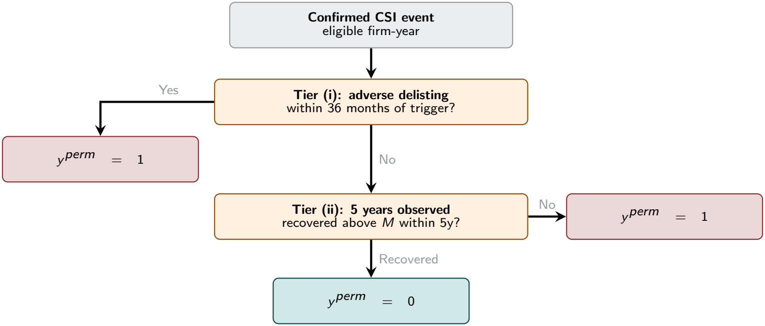


Buckets classify each CSI firm by itsfive-year post-classification performance. Light blue = the firm performs below the defined M threshold. Orange = the firm exceeds the M threshold, but does not recover entirely from the drawdown. Red = the firm recovers entirely from the drawdown.

Methodology III: "Permanent-CSI"

Dealing with recovering firms

To deal with the prior outlined false-positives due to observing a non-neglectible portion of recovering firms, it is proposed to test a secondary classification methodology: "Permanent-CSI". This methodology is even further forward looking, if a firm did not recover above the defined M threshold after 5 years, it will be classified as a CSI-event.



Applying the classification: Permanent CSI

By construction, Permanent-CSI is more restrictive than the initial classification methodology. The whole datasets now records 6,258 responses with $y = 1$ (compared to 8,517 originally). The prevalence drops from 4,74 to 3,34%. Overall, the prevalence rate is similar amongst the train and test set, with the test set showing a slightly smaller prevalence of the response variable.

Split	Obs. rows	$y = 0$	$y = 1$	Labelled	Prev.
Full	188,460	181,368	**6,258**	187,626	3.34%
Train	143,173	137,794	5,379	143,173	3.76%
Test	18,111	17,511	542	18,053	3.00%
OOS	27,176	26,063	337	26,400	1.28%

Low prevalence in the OOS set

The OOS dataset shows a markedly smaller prevalence rate of the response variable. The reason being that the forward-looking classification-methodology is constrained by the available data periods. Yet, this is not an issue as one does not require labelling the OOS dataset, which will be used to mainly evaluate the index-construction performance at the later part of the thesis.

Dataset: Descriptive Statistics

CSI positives are rare and economically distinct from non-positives

Both label definitions isolate a small distressed tail. The key contrast is not only prevalence: positive firms are much smaller and show worse realized returns and profitability.

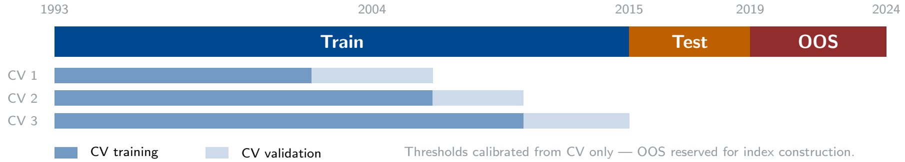
Sample	CSI	y	Obs.	Share	Firms	Median mcap	Ann. ret.	ROA
Full	Temp.	0	171,269	95.26%	19,073	254.9	4.9%	1.6%
Full	Temp.	1	**8,517**	4.74%	5,603	**74.9**	**-35.0%**	**-18.8%**
Full	Perm.	0	181,368	96.66%	19,565	252.6	4.0%	1.5%
Full	Perm.	1	6,258	3.34%	4,696	**65.7**	**-37.5%**	**-20.0%**

Remark: Market cap is USD millions.

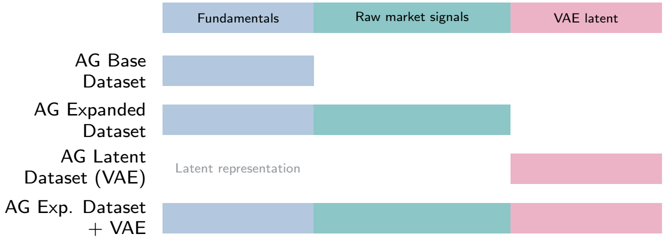
Even though the classification methodology seems to seperate firms rather well, it also constrains the expectations for the index-construction part of this thesis. CSI-classified firms are considerably smaller (measured by market cap.) than their peers. Thus, their index weight will be smaller, which implies only a minor minor contribution to overall index-returns. Consequently, a priori one would not expect to observe a drastic outperformance by avoiding or downweighting CSI-classified firms.

Modelling I: Setup and Feature Engineering

Validation design



Final feature-set keys



Modelling II: Results Temporary CSI

Description

Temporary CSI is a rare-event classification problem: only 4.74% of labelled firm-years are positives. Because of this class imbalance, AP and fixed-FPR recall are more decision-relevant than AUC alone. AG Exp. Dataset + VAE ranks best in CV, while AG Expanded Dataset is strongest on the 2016-2019 test split.

Model	CV-AP	CV-AUC	CV-FPR3	Test-AP	Test-AUC	Test-FPR3
AG Expanded Dataset	0.2046	0.8635	0.2344	**0.1985**	**0.8766**	**0.2002**
AG Base Dataset	0.1923	0.8446	0.2206	0.1656	0.8538	0.1517
AG Latent Dataset (VAE)	0.1659	0.8454	0.1706	0.1501	0.8438	0.1294
AG Exp. Dataset + VAE	**0.2114**	**0.8667**	**0.2478**	0.1864	0.8741	0.1779

Remark: CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A10.

Interpretation

Test-AP improves from the 4.44% test prevalence baseline to 19.85%, showing meaningful concentration of CSI cases near the top of the risk ranking. Test-AUC of 0.8766 confirms strong broad ranking, but AP is the more useful indicator for practical screening.

Modelling III: Results Permanent-CSI

Description

Permanent CSI is the harder task: positives are rarer than temporary CSI, with only 3.34% positives in the full labelled sample and 3% in the test split. Despite this, the models still rank risk well. AG Expanded Dataset remains the reporting baseline, while AG Exp. Dataset + VAE is the strongest non-raw challenger on CV/test.

Model	CV-AP	CV-AUC	CV-FPR3	Test-AP	Test-AUC	Test-FPR3
AG Expanded Dataset	0.1854	**0.8757**	**0.2607**	**0.1416**	0.8810	0.1808
AG Base Dataset	0.1785	0.8677	0.2379	0.1289	0.8627	0.1697
AG Latent Dataset (VAE)	0.1426	0.8460	0.1960	0.1115	0.8496	0.1513
AG Exp. Dataset + VAE	**0.1883**	0.8719	0.2578	0.1415	**0.8838**	**0.2011**

Remark: CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A11.

Interpretation

Temporary CSI has more positives and higher test prevalence: 4.44% vs 3%. Permanent CSI is therefore harder and more affected by class imbalance. AUC is similar or slightly higher for permanent CSI, but AP is lower because the base prevalence is lower. AG Expanded Dataset and AG Exp. Dataset + VAE are nearly tied on Test-AP: 0.1416 vs 0.1415. AG Exp. Dataset + VAE has the best Test-AUC and Test-FPR3, so it is a strong challenger, but not a clean replacement for the reporting baseline.

Modelling Summary: the Models Rank Implosion Risk Well

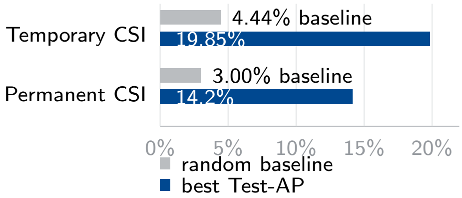
AutoGluon weighted ensemble

Expanding-window CV; thresholds from CV only

4 feature sets

Rare-event metrics: AP, recall@FPR

Test Average Precision vs random baseline (prevalence)



Result. Test-AUC is around 0.88, recall@FPR3 is around 0.20, and Test-AP is about 4.5–4.7× the base rate.

Verdict. AG Expanded Dataset is the reporting baseline; AG Exp. Dataset + VAE is the strongest challenger. Weighted ensembles top the leaderboards, and tree families carry the useful signal.

Subquestions. VAE augments rather than replaces the expanded feature set; feature drivers move to the appendix.

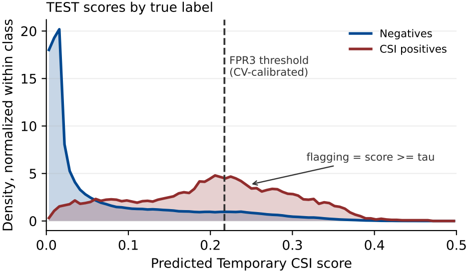
Hand-off

Calibrated scores become exclusion signals for the index; the question shifts from "can we rank risk?" to "does removing flagged firms improve the index?"

Remark: Model metrics and caveats are in Appendix A10–A13.

Why the Screen Cannot Be Perfect

Temporary CSI, AG Expanded Dataset: FPR3 threshold calibrated on CV predictions only, then applied to the 2016–2019 test set.



Live test diagnostic

- CV-calibrated $\tau = 0.217$; test FPR is 8.7% and recall is 45.9%.
- The rule flags 369 true CSI rows and 1,502 negatives in test.
- Only 10.8% of false positives are threshold-adjacent.
- 100.0% of false positives overlap the flagged true-positive score range.

Raising the threshold would remove some false positives, but it would also drop real CSI firms.

What this means for the index

The screen is useful but structurally imperfect. CSI firms are a small cap-weighted exposure, so false positives and retained-stock reweighting create a dilution ceiling. This is not simply a model failure; it frames the attribution slide.

Remark: Threshold is CV-calibrated; distribution and counts use the 2016–2019 test set; AG Expanded Dataset; temporary CSI. Threshold-adjacent means a false-positive score in $[\tau, \tau + 0.01]$, matching the prior false-positive diagnostic.

Modelling V: VAE Features — Benefits and Drawbacks

Autoencoder/AP subquestion. How does the integration of Autoencoders for feature engineering impact the Average Precision of Ensemble models compared to those trained solely on raw financial data?

Benefits

- Among VAE/non-raw variants, AG Exp. Dataset + VAE leads CV-AP in both tracks: temporary 0.2114, permanent 0.1883.
- Temporary OOS: AG Exp. Dataset + VAE leads AP 0.3152; R@FPR1/3/5 is 0.1051 / 0.2632 / 0.4128.
- Permanent test: AG Exp. Dataset + VAE leads AUC 0.8838 and FPR3/5 recall 0.2011 / 0.3155.
- Permanent OOS: AG Latent Dataset (VAE) leads AP/AUC/Brier/FPR recall: AP 0.0482, AUC 0.8337.

Drawbacks

- VAE does not replace expanded-data information.
- AG Expanded Dataset keeps slightly higher temporary OOS AUC: 0.8961 vs. 0.8950.
- AG Expanded Dataset is the cleaner reporting baseline.
- Latent-only is weaker on temporary CV/test and mainly competitive for permanent OOS.
- Added architecture and compute cost.

Verdict

Autoencoder features improve Average Precision as an augmentation, especially when combined with expanded features, but do not dominate AUC and do not replace raw/fundamental information.

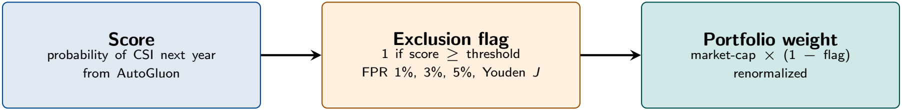
Remark: Values use current model-suite threshold metrics: `03_Data_Output/6_ModelSuite/derived_metrics/complete_threshold_metrics_wide.csv`; raw comparator metrics use `raw/metric_snapshot.csv`.

Index Construction I: From Score to Portfolio Weight

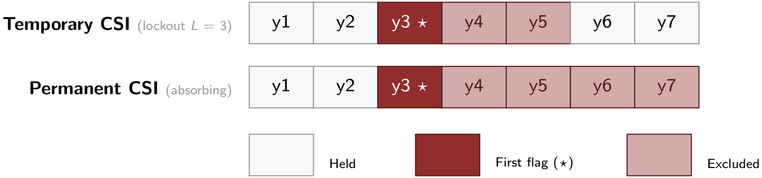
One score rule, two exclusion regimes

The model produces a probability of CSI next year. A threshold turns that probability into a binary exclusion flag. The flag is then applied to market-cap weights. Two regimes differ only in how long the flag persists after it first fires.

From model score to portfolio weight



How the flag evolves over time — a firm crosses the threshold in year 3



Index Construction II: Universes and Comparators

Same exclusion rule, four universes

The calibrated score-to-weight pipeline is applied independently inside each universe, so each screened index is evaluated against the matching unfiltered benchmark.

Total Market	**Large Cap**	**Mid Cap**	**Small Cap**
all eligible firms $\geq$ USD 100M	prior-Dec. rank top tier	prior-Dec. rank middle tier	prior-Dec. rank bottom tier

Tier breakpoints: <author to insert exact Large/Mid/Small cap thresholds>

Universe construction

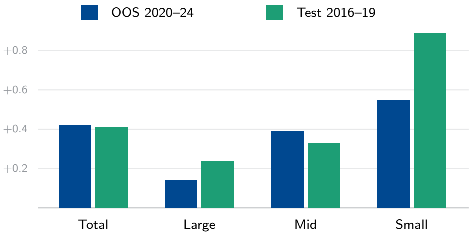
- Total Market keeps all eligible firms above USD 100 million market cap.
- Large, Mid, and Small Cap are formed by prior-December market-cap rank.
- Quarterly rebalancing: March, June, September, December.
- Within each universe, portfolio weights start from market capitalization.

Comparators

- Primary comparator: the unfiltered cap-weighted benchmark in the same universe.
- Screened portfolios are therefore compared like-for-like: Total vs Total, Large vs Large, Mid vs Mid, Small vs Small.
- Min-volatility and quality screens are planned extensions under the main research question; see limitations.

Index Results: Does the Crash Filter Help?

Benchmark-relative alpha by universe (pp, 10 bps net)



OOS alpha. Current OOS deltas are positive in all four universes; Mid and Small are strongest.

Test-set check. Test evidence is shown separately; Permanent-CSI Total Sharpe rises $0.89 \rightarrow \mathbf{1.00}$.

Source of alpha. Mostly retained-name reweighting, not direct event-avoidance (next slides).

Summary

The crash-filter appears beneficial in both evaluation windows, but the evidence differs by split. The OOS result is a positive benchmark-relative return delta across the four universes. The test-set result is reported separately and should not be read as the same diagnostic as OOS risk performance.

Temporary-CSI Results: OOS (2020-2024) with 10 bps of Transaction Costs

Best strategy per universe net of 10 bps costs vs. the zero-cost market-weighted benchmark.

		Return		Risk	
Universe	Strategy	Geo ret.	Δ pp	Sharpe	Max DD
Total	AG Base Dataset; Youden 3y	13.3 $\rightarrow$ 13.7	+0.42	0.60 $\rightarrow$ **0.64**	$-25.0 \rightarrow$ **-23.8**
Large	AG Base Dataset; Youden 3y	14.3 $\rightarrow$ 14.4	+0.14	0.66 $\rightarrow$ **0.68**	$-24.8 \rightarrow$ **-23.9**
Mid	AG Base Dataset; FPR5 5y	9.7 $\rightarrow$ 10.0	+0.39	0.41 $\rightarrow$ **0.42**	$-24.6 \rightarrow$ **-24.5**
Small	AG Exp. Dataset + VAE; Youden 3y	7.7 $\rightarrow$ 8.2	**+0.55**	0.31 $\rightarrow$ **0.33**	$-29.1 \rightarrow -29.3$

Main takeaway

The risk-filter works better the smaller the market cap.: the alpha rises from +0.14pp (Large) to +0.39pp (Mid) to +0.55pp (Small). Smaller-cap universes include more firms in distress and exclusions move a larger part of the portfolio (about 85-106% turnover vs. 10-13% for the total market). Drawdown improves everywhere except for small caps.

Permanent-CSI Results: OOS (2020-2024) with 10 bps of Transaction Costs

Best strategy per universe net of 10 bps costs vs. the zero-cost market-weighted benchmark.

		Return		Risk	
Universe	Strategy	Geo ret.	Δ pp	Sharpe	Max DD
Total	AG Exp. Dataset + VAE; FPR5	13.3 → 13.5	+0.26	0.60 → **0.61**	−25.0 → **−24.7**
Large	AG Exp. Dataset + VAE; FPR5	14.3 → 14.5	+0.22	0.66 → **0.67**	−24.8 → **−24.5**
Mid	AG Base Dataset; FPR5	9.7 → 10.3	**+0.62**	0.41 → **0.44**	−24.6 → **−24.5**
Small	AG Latent Dataset (VAE); FPR3	7.7 → 7.9	+0.24	0.31 → **0.32**	−29.1 → −29.3

Main takeaway

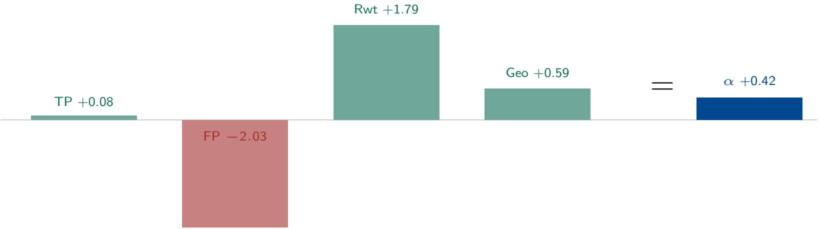
The permanent filter tilts the gains toward Mid Cap (+0.62pp), its strongest universe. Direction matches the temporary track, but more exclusion over-removes firms with broad thresholds, which is why Youden is dropped in favour of FPR3 and FPR5.

Where Does the Alpha Come From?

It is not direct event-avoidance

Because positives are rare, realized alpha is dominated by reweighting retained firms, not by directly dodging CSI events. A large false-positive opportunity cost is the main offset. For this reason, most of the performance gain disappears.

Decomposition for Total Market, OOS (pp): TP + FP + reweight + cost + geo = alpha



Universe (OOS, temp)	TP	FP	Reweight	TC	Geo	Alpha
Total	+0.08	−2.03	+1.79	−0.01	+0.59	**+0.42**
Large	+0.00	−1.50	+1.24	−0.01	+0.41	**+0.14**
Mid	−0.00	+0.05	+0.27	−0.11	+0.18	**+0.39**
Small	+0.42	−4.00	+3.94	−0.09	+0.28	**+0.55**

Permanent-CSI Results: Test (2016-2019) with 10 bps of Transaction Costs

		Return		Risk	
Universe	Strategy	Geo ret.	Δ pp	Sharpe	Max DD
Total	AG Exp. Dataset + VAE; Youden	13.8 → 14.2	+0.45	0.89 → **1.00**	−14.6 → **−12.8**
Large	AG Exp. Dataset + VAE; Youden	14.2 → 14.4	+0.26	0.95 → **1.04**	−13.8 → **−12.2**
Mid	AG Exp. Dataset + VAE; Youden	13.7 → 14.2	+0.43	0.84 → **0.95**	−15.3 → **−13.8**
Small	AG Exp. Dataset + VAE; Youden	11.4 → 12.2	**+0.83**	0.58 → **0.68**	−20.1 → **−18.3**

Validation Summary

In the calmer 2016-2019 window a single consistent strategy (Exp. Dataset + VAE with Youden computed thresholds) improves Sharpe and cuts drawdown in every universe, with Total Sharpe crossing 1.0. The same small-cap tilt appears (+0.83pp Small vs. +0.26pp Large).

Impact of transaction costs: the winners are unchanged

Selected OOS winners when assuming transaction costs of 5, 10 and 20 bps.

Track	Universe	Selected OOS winner	Active α (5/10/20)	Ranking
Temp.	Total	AG Base Dataset; Youden 3y	$+0.43/+0.42/+0.41$	unchanged
Temp.	Large	AG Base Dataset; Youden 3y	$+0.15/+0.14/+0.12$	unchanged
Temp.	Mid	AG Base Dataset; FPR5 5y	$+0.45/+0.39/+0.28$	unchanged
Temp.	Small	AG Exp. Dataset + VAE; Youden 3y	$+0.59/+0.55/+0.45$	unchanged
Perm.	Total	AG Exp. Dataset + VAE; FPR5	$+0.26/+0.26/+0.25$	unchanged
Perm.	Large	AG Exp. Dataset + VAE; FPR5	$+0.22/+0.22/+0.20$	unchanged
Perm.	Mid	AG Base Dataset; FPR5	$+0.68/+0.62/+0.51$	unchanged
Perm.	Small	AG Latent Dataset (VAE); FPR3	$+0.28/+0.24/+0.16$	unchanged

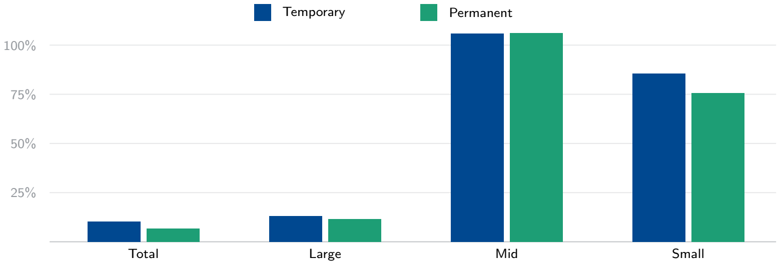
Takeaway

Raising costs to 20 bps lowers net returns but does not change a single winner or flip the strategy ranking.

Remark: Active alpha is measured versus the zero-cost market-cap benchmark; only the strategy pays transaction costs.

Cost Drag due to turnover is concentrated in mid and small caps

Annualized gross buy + sell turnover of the selected OOS strategies (%)



Summary

Drag scales linearly with turnover × cost. Total and Large are barely traded (between 7-13%), so their 20 bps drag is at most 0.13pp. Mid and Small rebalancing is comparably larger at 75-106% a year, lifting their 20 bps drag to roughly 0.76-1.06pp. Yet, the winner ranking on the previous slide still holds.

The Selected Rules Are Principled, Not Grid-Mined

Temporary CSI — finite lockout

Youden 3yr wins Total, Large, and Small; FPR5 5yr wins Mid. A recoverable 3–5yr lockout lets the signal age out, capping turnover outside Mid / Small.

20 bps evidence: Youden has the strongest mean best alpha, +0.24pp.

Permanent CSI — strict FPR

FPR5 wins Total, Large, and Mid; FPR3 wins Small. Absorbing removal needs narrower gates — broad Youden over-removes names and compounds turnover.

20 bps evidence: FPR3 / FPR5 lead (+0.24 / +0.22pp); Youden averages −0.31pp.

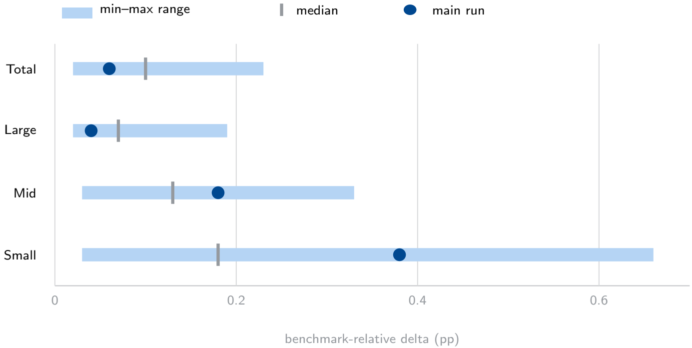
Why this matters

The two tracks select different rule families for a structural reason, not by grid-mining: a temporary flag can expire, so it tolerates a looser cutoff; a permanent flag is irreversible, so cost and turnover mistakes must be gated more tightly. Turnover detail is on the previous slide.

Remark: Source: `threshold_family_summary_20bps.csv`; `best_by_track_index_cost.csv`.

Sensitivity Analysis: temporary CSI only

Main run versus 24 completed temporary-CSI C/M/T sensitivity runs; relative geometric-return delta (pp).



Robustness read

All completed temporary-CSI C/M/T runs stay positive; sensitivity changes magnitude, not sign. Main run: below median in Total/Large, above in Mid/Small. Temporary-CSI evidence only.

Remark: Temporary CSI only; three blocked-partial configs excluded. Source: `universe_stability_summary.csv`.

Appendix A1: Dataset, Labels, and Detection Checks

Failure-label audit

Signal	Firms	Detected	Rate
CRSP 572–574	629	545	86.65%
CRSP 574 only	548	513	93.61%
Adverse 400–490, 550–585	5,654	3,676	65.02%

Temporary CSI has 8,517 positive rows and 8,674 unresolved rows. Permanent CSI has 6,258 positive rows and 834 unresolved rows.

Clean labelled prevalence

Track	Split	Labelled	Pos.	Prev.
Temp.	Train	143,173	6,543	4.57%
Temp.	Test	18,111	804	4.44%
Temp.	OOS	18,502	1,170	6.32%
Perm.	Train	143,173	5,379	3.76%
Perm.	Test	18,053	542	3.00%
Perm.	OOS	26,400	337	1.28%

Read

The labels stay rare after the observable-scaffold cleanup. CRSP terminal-failure information is an additive audit/override around the CSI timing rule, not an independent default label.

Source: Sources: AE-PANEL label-count and CRSP-overlap evidence used in the original appendix source map.

Appendix A2: Descriptive Statistics and Feature Groups

Positive firms look distressed before modelling

Track	Class	Mkt cap	Ann. ret.	ROA
Temp.	y=0	350.9	4.9%	1.6%
Temp.	y=1	68.7	-35.0%	-18.8%
Perm.	y=0	249.3	4.2%	1.5%
Perm.	y=1	66.2	-37.5%	-19.3%

Market cap is the full-sample median in USD millions.

Feature blocks retained

- Point-in-time accounting ratios: leverage, liquidity, profitability, accruals, cash flow, and Altman-style distress.
- Firm histories: differences, accelerations, expanding means/volatility, rolling minima/maxima/trends, and peak/trough moves.
- Price and macro signals: momentum, volatility, drawdowns, macro levels, and selected firm $\times$ macro interactions.
- Feature sets: base, expanded, VAE latent, and expanded plus VAE.

Source: Sources: descriptive-statistics CSVs and feature-dictionary evidence from the original appendix/source map.

Appendix A3: CV Design and Model Metrics

Expanding-window CV

Fold	Training	Validation
1	Initial block	seed only
2	through 1998	1999–2004
3	through 2004	2005–2009
4	through 2009	2010–2015

Split-before-preprocessing discipline is retained: winsorization, imputation, PIT scaling, and thresholds are fitted on training/CV information only.

Payoff metrics used in the main story

Track	Split/model	AP	AUC	R@FPR3
Temp.	Test AG Exp.	0.1985	0.8766	0.2002
Temp.	OOS AG +VAE	0.3152	0.8950	0.2632
Perm.	Test AG Exp.	0.1416	0.8810	0.1808
Perm.	Test AG +VAE	0.1415	0.8838	0.2011
Perm.	OOS VAE latent	0.0482	0.8337	0.1484

Metric interpretation

AP and fixed-FPR recall are emphasized because positives are rare. Economic claims are made only after thresholding, exclusions, turnover, and returns in the index section.

Source: Sources: `03_Data_Output/6_ModelSuite/comparison/AE-MODEL-SUITE-007_model_suite_metrics_long.csv` and raw derived metrics.

Appendix A4: VAE and Model-Family Evidence

VAE setup

Item	Setting
Encoder	$n \to 256 \to 128 \to 64$
Latent dim.	24
Decoder	$z \to 64 \to 128 \to 256 \to n$
Loss	Recon. $+ \beta$ KL $+ \gamma$ CSI loss
Weights	$\beta = 1.0$, $\gamma = 0.1$

Test and OOS labels are not used for VAE fitting.

Model-family result

- All eight final track $\times$ feature-set AutoGluon leaderboards select `WeightedEnsemble_L2`.
- Useful base signal is mostly tree/boosted-tree based; neural nets were tried but do not dominate.
- AG Expanded is the reporting baseline; AG Expanded + VAE is the strongest challenger in the main modelling story.
- VAE augments the expanded predictors; it does not replace tabular fundamentals.

Source: Sources: model-suite family-winner CSVs, compact hyperparameter JSONs, ensemble-weight JSONs, and VAE evidence files.

Appendix A5: Index Grid and Selected Rules

Completed economic grid

- Tracks: temporary CSI and permanent CSI.
- Universes: Total, Large, Mid, and Small.
- Models: AG Expanded, AG Base, AG Latent (VAE), and AG Expanded + VAE.
- Thresholds: Youden, FPR1, FPR3, and FPR5.
- Transaction costs: 0, 5, 10, and 20 bps on strategy turnover.

Rules used in the final selected strategies

- Temporary CSI uses finite exclusion lockouts: 1, 2, 3, or 5 years.
- Permanent CSI is absorbing after the first active signal.
- Temporary winners mostly use Youden with longer finite lockouts.
- Permanent winners mostly use FPR3/FPR5, where the absorbing rule makes false positives more costly.

Interpretation

The rules translate calibrated risk ranks into implementable exclusions. Model quality is necessary, but realized alpha also depends on event timing, universe weights, turnover, and benchmark returns.

Source: Sources: `full_grid_manifest.csv`, `best_by_track_index_cost.csv`, threshold-family summaries, and turnover summaries.

Appendix A6: Transaction Costs and Attribution Sources

20 bps robustness summary

Track	Univ.	Alpha 0	Alpha 20
Temp.	Total	+0.43	+0.41
Temp.	Large	+0.15	+0.12
Temp.	Mid	+0.51	+0.28
Temp.	Small	+0.64	+0.45
Perm.	Total	+0.27	+0.25
Perm.	Large	+0.23	+0.20
Perm.	Mid	+0.74	+0.51
Perm.	Small	+0.32	+0.16

Attribution/source paths

- Headline winners and TC overlays come from AE-INDEX-SUITE final tables and comparison outputs.
- Contribution slides use config-level attribution plus model-specific error-cost decompositions.
- Test-set contribution rows use main-suite `period=test` attribution where isolated test diagnostics are unavailable.
- Active alpha is measured versus the zero-cost market-cap benchmark; only the strategy pays transaction costs.

Source: Sources: `03_Data_Output/7_IndexConstructionValidation/nonraw_index_suite/` comparison/final-table outputs and AE-ATTRIB evidence.

Appendix A7: Temporary CSI Sensitivity Detail

Role	Run ID	Pos.	OOS AP	OOS AUC	11C alpha	Status
Composite	C090_M000_T012	12,761	0.557	0.920	+0.169	complete
AP winner	C060_M000_T012	31,596	0.566	0.864	+0.052	reused
11C winner	C090_M020_T018	6,035	0.274	0.911	+0.226	complete
Baseline	C080_M020_T018	8,517	0.308	0.896	+0.065	reused

Main-run comparison and limits

The accepted main run C080_M020_T018 is the continuity baseline, not the top sensitivity setting. Across completed/reused temporary-CSI C/M/T runs, sensitivity changes magnitude more than direction. This sensitivity evidence is temporary-CSI only and does not support permanent-CSI sensitivity claims.

Incomplete grid disclosure

Three configurations remain `blocked_partial`: C080_M000_T012, C080_M000_T018, and C060_M020_T028.

Source: Sources: local AE-SENS summary CSVs and temporary transaction-cost overlay files.

Appendix A8: Limitations, Future Work, and Source Traceability

Limitations kept explicit

- Sensitivity evidence is temporary-CSI only.
- Test-set attribution caveats are documented where main-suite `period=test` rows are used.
- Model metrics do not establish index alpha by themselves.
- Feature-importance slides are directional perturbation evidence, not full AutoGluon-native importance.

Traceability pointer

- Dataset/model/index/sensitivity claims map to source files in the slide-source map and ticket evidence.
- Conceptual frames and future-work notes are marked as non-result claims.
- Future robustness: permanent-CSI sensitivity, low-vol/quality benchmarks, probability weighting, and terminal-failure miss decomposition.

Source-map status

This cleaned Draft appendix is a presentation-compression pass. The ticket evidence records the appendix remapping and the feature-importance source files used for A9–A10.

Appendix A9: Feature Importance – 11 Families

Temporary CSI, AG Expanded family-level perturbation; top canonical families

Family	Abs. response	Signed	N	Scale
Point-in-time ratios	0.609	-0.121	53	
Rolling-window statistics	0.523	-0.083	180	
Peak deterioration	0.152	-0.021	18	
Expanding volatility	0.098	+0.005	38	
YoY changes	0.094	-0.015	38	
Expanding mean	0.065	-0.007	38	
Trough rise	0.048	+0.002	7	
Price/macro interactions	0.039	-0.004	5	
Acceleration	0.037	-0.002	38	
Accounting momentum	0.037	-0.004	15	
Consecutive declines	0.008	+0.001	10	

Caveat

Model-based log-odds perturbation on bounded GBM-only predictor workspace; CV/training rows only; interpret as directional model-response evidence, not full AutoGluon-native feature importance.

Source: `03_Data_Output/10_FeatureImportance/family_importance/AE-FEAT-IMPORT-004R_family_importance_summary.csv`.

Appendix A10: Feature Importance – Ratios and Individual Drivers

Point-in-time ratios

Ratio	Abs.	Signed
earn_yld	0.133	-0.017
altman_z2	0.131	+0.012
ocf_per_share	0.101	-0.015
roic	0.067	-0.019
roa	0.048	-0.020
ocf_margin	0.046	-0.009
interest_cov	0.042	-0.013
quick_ratio	0.028	-0.003

Individual features

Feature	Abs.	Family
unrate	0.181	PIT ratios
earn_yld	0.133	PIT ratios
altman_z2	0.131	PIT ratios
ocf_per_share	0.101	PIT ratios
peak_drop_log_mkvalt	0.096	Peak drop
roll_min_5y_earn_yld	0.086	Rolling
roll_min_3y_earn_yld	0.067	Rolling
roll_sd_5y_earn_yld	0.067	Rolling

Caveat

Model-based log-odds perturbation on bounded GBM-only predictor workspace; CV/training rows only; interpret as directional model-response evidence, not full AutoGluon-native feature importance.

Source: AE-FEAT-IMPORT-005R_pit_ratio_importance_summary.csv; AE-FEAT-IMPORT-006R_individual_feature_importance_summary.csv.